Ahmad Atyani

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Nottingham, UK • Open to relocation

PROFILE

Electronics and Computer Engineering graduate (First Class Honours, 78%, University of Nottingham) focused on embedded firmware and electronics hardware. Writes C and C++ for real-time acquisition and control on STM32 and ESP32, bare-metal and under Zephyr RTOS, and designs, fabricates and brings up custom PCBs in KiCad. Takes analogue and RF circuits from specification through simulation and tolerance analysis to measurement. Industry experience in driver development, embedded validation and board-level fault finding at Exalt Technologies.

EDUCATION

BEng Electronics and Computer Engineering, University of Nottingham Sep 2023 to Jun 2026
First Class Honours, 78%. Modules include embedded systems, analogue electronics, RF and microwave engineering, signal processing and communications, control systems, digital and CPLD design.

International Baccalaureate, Friends School, Ramallah *42/45.* Jul 2022

WORK EXPERIENCE

Embedded Systems Intern, Exalt Technologies Jul to Sep 2025

- Implemented and compared I3C and I2C protocol drivers on STM32 hardware, measured behaviour against the expected specification, and wrote the technical recommendation used to support protocol selection.
- Developed and debugged embedded C applications on STM32 Nucleo boards (GPIO, PWM, interrupts, real-time scheduling) under Zephyr RTOS and bare-metal, working in Linux-based toolchains.
- Performed testing, validation and fault finding on multi-board Ethernet applications, reproducing and isolating defects across board configuration, firmware and networking, in a cross-functional team of software and hardware engineers.

FINAL-YEAR INDIVIDUAL PROJECT

Automated 3D Scanning and Illumination Rig for Surgical Visualisation, University of Nottingham 2025 to 2026
Solo ownership of an imaging product on a £200 budget cap: requirements, firmware, electronics, mechanical design and validation.

- Wrote ESP32 firmware in C++ for real-time LED and stepper-motor control and integrated it with a dual-camera capture pipeline on Raspberry Pi 5 (Linux); documented design decisions and trade-offs in a 61-page dissertation.
- Designed a custom LED ring PCB in KiCad from schematic capture through layout, hand-soldered SMD and through-hole parts, then brought up and debugged the board with bench power supply, function generator and oscilloscope.
- Diagnosed a motor-driver fault through systematic board-level fault finding, resolving it by selecting and integrating an A4988 driver in place of an L298N; designed gears and mounts in Fusion 360.
- Derived test criteria from a 9-point requirements specification (passed 7, 2 partial, no fails) and built an automated Python and OpenCV harness reporting 93% inter-image overlap and 0.906 to 0.913 illumination uniformity.

ENGINEERING PROJECTS

Formula Student, VCU Software and Hardware Sub-Teams 2024 to 2025

- Wrote STM32 firmware in STM32CubeIDE handling CAN bus messaging between vehicle subsystems, working to the team's message definitions and timing requirements.
- Designed a safety-critical 60 V indicator circuit to Formula Student regulations, including component selection and compliance against the rulebook.

Pulse Oximeter, Mixed-Signal Analogue Front End and Firmware 2024

- Built an optical SpO2 and heart-rate system end to end: analogue front end for the photodiode signal chain, FFT-based processing to recover heart rate, STM32 firmware in C and C++ using on-chip ADC and DMA acquisition, an RS-485 transceiver link for board-to-PC data transfer, and CPLD logic in Quartus Prime.

ACADEMIC COURSEWORK PORTFOLIO

- **Low-Pass Microstrip Filter Design (Microwave Engineering coursework):** designed a 4th-order Butterworth prototype to a 3 GHz cutoff and 14 dB at 5 GHz insertion-loss specification, verified 17.8 dB in Keysight ADS through S11 and S21 analysis, then synthesised microstrip layouts in LineCalc and compared a Kuroda-identity implementation against a stepped-impedance version on footprint and stopband rejection.
- **Active and Digital Filter Design (Signal Processing coursework):** designed a 4th-order Butterworth Sallen-Key low-pass filter to a passband and attenuation specification; LTspice measured a 133.88 Hz cutoff against a 134 Hz target and 45.83 dB at 500 Hz, then ran a component tolerance sensitivity analysis and built the digital counterpart through z-domain pole-zero placement.
- **Two-Stage BJT Common-Emitter Amplifier (Analogue Electronics coursework):** sized the bias network for a 2 mA quiescent collector current, predicted a midband gain of 50.6, reconciled hand calculations against LTspice, applied Miller's theorem to the high-frequency corner, and used output FFTs to show harmonic distortion emerging outside the linear region.

TECHNICAL SKILLS

- **Embedded and Microcontrollers:** STM32 (bare-metal and Zephyr RTOS), ESP32, Arduino, Raspberry Pi (Linux), device drivers, interrupts, GPIO and PWM, real-time scheduling, ADC and DMA acquisition, STM32CubeIDE
- **Interfaces and Protocols:** I2C, I3C, SPI, UART and serial, CAN bus, RS-485, TCP and UDP over Ethernet
- **Hardware and PCB:** KiCad (schematic capture and layout), SMD and through-hole soldering, hardware bring-up and debug with oscilloscope, function generator and bench supply, reading schematics and datasheets, Fusion 360
- **Analogue and Signal Processing:** Analogue signal chain and active filter design (Butterworth, Sallen-Key), BJT amplifier design and small-signal analysis, component tolerance analysis, digital filter design, FFT-based processing, sampling, SNR and BER characterisation, LTspice, MATLAB, Simulink
- **RF, Digital and Control:** Keysight ADS, LineCalc, S-parameter analysis, impedance matching, microstrip filter synthesis; Verilog (simulation) and CPLD design in Quartus Prime; transfer function modelling, root locus, Bode analysis, PI controller design
- **Programming and Tools:** C, C++ (C++17, object-oriented design, Qt), Python (NumPy, Pandas, OpenCV, PyTorch), MATLAB; Git (branches, pull requests, code reviews), GitHub Actions CI, CMake, MSVC and Linux toolchains, Docker, unit and integration testing, requirements specifications and test plans

CERTIFICATIONS, LANGUAGES AND INTERESTS

- **Certifications:** Harvard CS50; MATLAB Onramp; Google Data Analytics Professional Certificate; Kaggle Python. **Languages:** Arabic (native), English (fluent), German (A2).
- **Activities:** Formula Student team member; personal embedded and PCB projects; competitive programming practice in C++.